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Can Artificial Intelligence Improve Gender Equality? Evidence from a Natural Experiment

Leo Bao,[a] Difang Huang,[b,*] Chen Lin[c]

[a] Department of Banking and Finance, Monash Business School, Monash University, Caulfield East, Victoria 3145, Australia; [b] Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China; [c] Faculty of Business and Economics, University of Hong Kong, Hong Kong

*Corresponding author

Contact: Leo.Bao@monash.edu, https://orcid.org/0000-0002-4337-6919 (LB); difang@amss.ac.cn, https://orcid.org/0000-0002-1159-3134 (DH); chenlin1@hku.hk, https://orcid.org/0000-0003-4205-8633 (CL)

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Abstract. Gender discrimination in education hinders women's representation in various fields. How can we create a gender-neutral learning environment when teachers' gender composition and mindset are slow to change? Recent development in artificial intelligence (AI) provides a way to achieve this goal as engineers can make AI trainers gender neutral and not take gender-related information as input. We use data from a natural experiment in which such AI trainers replace some human teachers for a male-dominated strategic board game to test the effectiveness of AI training. The introduction of AI improves teaching outcomes for boys and girls and reduces the preexisting gender gap. Survey responses indicate that AI's information advantage, friendly appearance, and interactive features helped students to learn faster, and class recordings suggest that AI trainers' nondiscriminatory emotional status can explain the improvement in gender equality. We demonstrate AI's potential in improving learning outcomes and promoting diversity, equity, and inclusion in analogous settings.

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1. Introduction

Women are underrepresented in fields such as science, technology, engineering, and mathematics disciplines and strategic board games (e.g., *Go* and chess). For example, women only occupy less than 6% of seats in the most distinguished science positions in some developing countries nowadays (Bao and Huang 2022), and they are absent from the list of top 10 chess players from various rating systems at the beginning of 2022. Research documents various explanations for this phenomenon and finds that discriminatory practices and stereotypes are key contributors to the gender gaps (Alan et al. 2018, Lavy and Sand 2018, Carlana 2019). Gender discrimination can rise as early as primary education (see, e.g., Ganley et al. 2018, Breda et al. 2020); improving gender equality in education has been a critical social goal for a long time (e.g., Klasen 2002, Duflo 2012, Gallus and Heikensten 2020). Still, teachers and mentors can hold gender-biased beliefs, and these mentalities are sometimes slow to change. The development of artificial intelligence (AI) technology brings a possible solution to this problem. Nowadays, AI is already widely employed for educational purposes, and gender-related variables can be eliminated in the input to AI to improve gender neutrality. However, little evidence is documented on the effectiveness of AI training, especially its potential merit in reducing gender performance gaps.

We fill this vacancy by analyzing data from a natural experiment in which AI replaced a random subset of human teachers at a *Go* training agency because of quarantines against COVID-19. The field setting is ideal for comprehensive scrutiny of AI teaching for several reasons. First, students have to attend two training sessions per week, and each session consists of a tournament with an opponent with similar playing skills and a revision of the match with their teachers. Because of the simple training structure, we can study how a teacher's guidance in a revision can improve a student's *Go* skill in the coming games. The frequent playing also gives us a decent number of observations to evaluate the evolution of students' skill levels and make statistical inferences. Second, *Go* is a male-dominated game, and a gender performance gap exists before the observation window, allowing us to address the research question. Third, because of China's COVID prevention regulations, both tournaments and revisions were conducted online during the analysis period. The *Go*-playing software recorded all moves, enabling the quantification of student performance. We also recorded revision sessions (conducted by both human and AI teachers) to examine the relationship between teachers' facial, vocal, and verbal characteristics and student learning outcomes. It is important to highlight that the AI teacher is represented as a cartoon character on the screen, offering facial, vocal, and verbal information that can be analyzed, similar to human teachers. Moreover, the student–teacher interaction occurs online for both human and AI teachers, ensuring a fair comparison between the two teaching modes. Additionally, the AI does not gather any student information (e.g., gender) beyond the moves made in each tournament, thus fulfilling crucial requirements for this study.

We conduct a comprehensive analysis of tournament data for two groups: students taught solely by human teachers (control group) and students who received training from the AI teacher after the intervention (treated group). Prior to the introduction of AI training, the performance in *Go* exhibited a parallel trend for both groups. However, following the intervention, the treated group displayed a significantly faster improvement in *Go* skills compared with the control group. This indicates that the AI teacher surpasses human teachers in enhancing students' learning outcomes across the student population as a whole.

We then investigate how AI teaching affects boys and girls differently. We discovered a persistent gender performance gap prior to the experimental intervention; boys performed significantly better than girls. After the introduction of AI, we find both genders in the treated group made progress faster than their counterparts in the control group, and girls advance more rapidly than boys under AI training. Among the treated group, boys and girls achieved similar performance scores after five months of AI training.

In the next step, we surveyed students who experienced both AI and human teaching on teaching-related questions to understand potential channels that drive the treatment effects. Survey results show that AI's advantage in analyzing games, providing relevant statistics, and interactive features helped students to learn faster than those taught by human teachers; however, we failed to find any significant gender differences in responses to the above questions. We also discover that students tend to perceive the appearance of the AI teacher being more attractive than human teachers, and girls have a stronger preference for AI's appearance than boys. Students also claimed that the attractiveness of teachers is positively correlated with their learning outcomes.

We further analyzed video recordings of all revision classes to study why the gender performance gap existed in the first place and how AI training can reduce it. The data indicates that human teachers showed more positive and less negative emotions toward boys and students with more advanced *Go* skills. The data also finds significant correlations between teachers' emotional status in the revision class and students' performance in the subsequent games. However, the AI teacher's emotion had fewer variations than human teachers and was not gender dependent. Consistent with the video evidence, the survey results indicate that girls are capable of discerning gender-biased emotions exhibited by human teachers, which may influence their learning outcome. Nonetheless, the students did not perceive the AI teacher as exhibiting gender bias. Therefore, teachers' emotional status may explain the evolution of the gender performance gap.

We contribute to the literature on gender economics. In particular, we complement the literature studying the causes of gender gaps (Goldin et al. 2006, Niederle and Vesterlund 2010) and how to promote gender equality. We identify that human teachers' gender-biased emotions can lead to a gender performance gap in *Go* training, and the AI teacher can reduce such a gender gap. Compared with methods commonly applied in the field (e.g., affirmative actions, women-friendly institutional setups, sponsorship of women, and nudges/propaganda), we introduce a novel and low-cost solution to the gender disparity issue by employing AI. To the best of our knowledge, this study is the first to examine the merit of using AI to improve gender equality in the classroom. AI may reduce the gender gap by neutralizing the impact of discriminatory emotions from human teachers in other educational settings. However, we acknowledge that all training sessions in our study were held online; readers should proceed cautiously when extrapolating our findings to other teaching forms such as face-to-face (F2F) training.

Moreover, we provide novel evidence, with the help of AI, to understand interpersonal communication systematically. There is an emerging strand of literature that applies machine learning methods to analyze text messages, voice and vocal recordings, and images (Hoberg and Phillips 2010, Mayew and Venkatachalam 2012, Hu and Ma 2021). Our paper contributes to this strand of literature by using machine learning methods to process teaching videos. We quantify teachers' emotions in facial, vocal, and verbal dimensions and investigate how these emotions affect student behaviors.

Further, this paper adds to the literature examining the effects of AI adoption on significant socioeconomic outcomes (Acemoglu and Restrepo 2018, 2020; Bao et al. 2022). AI technology has become pervasive in almost all sectors, including the finance industry, legal system, medical and pharmaceutical innovations, and marketing. Patterns of discrimination toward women and racial minorities have been detected in algorithms used for advertisements, university admissions, legal decisions, and health management, among others (Santelices and Wilson 2010, Kleinberg et al. 2018, Baker and Hawn 2022). We extend this strand of literature by showing the potential influence of AI on education: AI teachers can improve teaching efficacy and gender equality with its information advantage and nondiscriminatory emotions.

Notably, decision making in games such as *Go* mirrors that of managers and policymakers, who employ intuition to solve complex problems under uncertainty and time constraints (Mnih et al. 2015, Miric et al. 2021). AI may possess an inherent advantage in analyzing such scenarios and providing relevant statistics with gender-neutral emotional statuses as AI technology can assist *Go* training. Additionally, AI excels in detecting discrimination and subtle emotional changes, thereby providing a means to mitigate bias and gather valuable information when participants' emotions manifest their thoughts during business negotiations, court hearings, and political discussions. Consequently, our findings have implications for AI's instructional role in enhancing decision making for managers and policymakers.

2. Background

2.1. The Game of *Go*

Go is an ancient, two-player, abstract strategy board game that originated in China more than 2,500 years ago. It is considered the oldest board game still played today with a global player base of more than 50 million people. Whereas most players reside in East Asia, the game's popularity is growing internationally with more than 50 professional tournaments held outside Asia each year.

The game is played on a 19×19 grid board with 361 crossing points. The objective for players is to enclose as many points as possible using black and white stones. The players take turns placing their stones on vacant points, starting with black stones. Once placed, the stones cannot be moved unless they are surrounded by the opposing player's stones on all connected points, resulting in capture and removal from the board. The game continues until both players decline further moves. The winner is determined by counting the surrounded territory with compensation (*komi*) added to the score of the player using white stones to offset the disadvantage of moving second.

Despite its simple rules, *Go* is highly complex, distinguishing itself from chess with a larger board that allows for more strategic possibilities and fewer move constraints. It boasts approximately 2.1×10^{170} legal stone positions, surpassing the number of atoms in the observable universe. Consequently, *Go* poses a formidable challenge for programmers (Silver et al. 2016).

Advancements in deep-learning algorithms have significantly enhanced *Go* programs (Silver et al. 2017). In notable victories, Google DeepMind's AlphaGo defeated European champion Fan Hui in 2015. It further shocked the *Go* community by defeating Ke Jie, the world No. 1 at the time, in 2017. These successes shattered the belief that computers were inferior in intuitive judgments amid immense complexity. AlphaGo's triumph changed the approach to studying *Go* with more players now incorporating AI technology into their training.[1]

Our AI training adopts a similar algorithm as AlphaGo, serving as a powerful alternative to human teachers. Besides evaluating moves for students as human teachers, AI shows the changes in the winning probability, providing more accurate and objective feedback. It also categorizes all moves into perfect, excellent, good, poor, error, and critical error categories based on the changes in the winning probability and produces a summary for both players during the game's opening, middle, and end.[2] For moves labeled error or critical error, the AI program provides a recommended choice and shows how the recommendation can improve the winning rate and the stone count. The AI then uses a face simulation program to construct a cartoon face and adopts text-to-speech technology to translate the review contents into voices for students. The cartoon image replaces the box that shows a human teacher's video image. The display of the *Go* revision software is similar for AI and human teachers. Compared with the screen of human teaching, the AI provides game statistics on separate screens and switches between these screens and the revision software. We plot the screenshots of AI teaching in Online Appendix E. However, because of the copyright constraint, we are unable to demonstrate screenshots of the cartoon figure.

2.2. Gender Disparity in *Go*

Gender disparity remains prevalent in *Go* playing with significantly fewer female amateur and professional players than males. According to an International *Go* Federation survey, women make up less than 20% of *Go* players in most countries. To encourage women's participation, various international *Go* associations have implemented measures to promote women's involvement in the game, including organizing women-only *Go* tournaments alongside major title tournaments.

Gender disparity is also evident in players' Elo ratings.[3] In May 2022, only 26 out of the top 400 professional players worldwide were female with an average Elo rating of 3,172.4, lower than that of male players at 3,286.2. The development of the *Go* game would benefit from the inclusion of more talented women players as they can bring new ideas, styles, and skills into play.

2.3. The Training of *Go*

Go players need systematic learning and intensive playing to improve their skills. In the training process, it is standard to let players with similar skills play together to make the game entertaining and avoid extreme inequality in the outcome. Therefore, the *Go* authority creates a hierarchical system to rank all players, facilitating training and evaluation. Players' performances in rank-grading tournaments determine their ranks.

The level of players is measured with *kyu* and *dan* ranks. *Kyu* ranks are student ranks, whereas *dan* ranks are generally considered master ranks. In China, beginners who start to learn the game acquire the 25th *kyu*. As they progress, they advance numerically downward through the *kyu* ranks. The best *kyu* rank attainable is, therefore, the first *kyu*. If players go beyond the first *kyu*, they receive the rank of first *dan* and move numerically upward through the *dan* ranks. Students in our sample advance in *dan* rank through a yearly rank-grading tournament during the summer holidays; thus, students' *dan* ranks did not change in the observation window (fall and spring semesters).

The *Go* training agency providing data for this study operates as the following. The training for students with *kyu* ranks includes consolidation of *Go* rules, learning tactics, practice in *tsumego* (*Go* puzzles representing critical situations that can only be saved by careful considerations), and study of *fuseki* (optimization in the placement of opening stones) and *joseki* (standard sequences of moves to play out when a specific situation happens).

The training for students with *dan* ranks includes two main parts, playing and revision. Teachers guide students to play *Go* with opponents from other training agencies across China. We contacted some of these *Go* training agencies and found that they were not using AI training at the time of the experiment; the opponents of our students were unlikely to be trained by AI. Furthermore, students were unable to communicate verbally or in writing during tournaments, making it difficult to determine whether their opponents had undergone AI training. Therefore, the opponents' training (AI or human teaching) is implausible to affect our main results.

All games are played through an online *Go*-playing software, and all moves are recorded. After each game, the teacher reviews this game with the student step by step in a one-on-one meeting to evaluate the strengths and weaknesses of the performance and provide suggestions for improvements. Replaying a game (i.e., a revision) is a common strategy that most *Go* masters employ to consolidate memory and improve strategical thinking.

In the COVID shock, some *kyu*-rank students' teachers were also quarantined. Instead of receiving AI training as for *dan*-rank students, their training became watching video recordings. Therefore, those *kyu*-rank students cannot inform us of the effectiveness of AI teaching.

3. Experiment Details

The natural experiment happened at one of the largest *Go* training agencies in China, and it has trained more than 10,000 pupils and promoted more than 20 students to enter professional competitions. Students were mostly schoolchildren, who were placed into different classes based on their *kyu*/*dan* at the beginning of the fall semester (i.e., September 2020), and the class was fixed across the experimental window. Because students with *kyu* ranks were not suitable to address the research question (as discussed above), we do not include them for experimental purposes; the discussion below only refers to students who had acquired *dan* ranks.

Students take two sessions every week; each includes playing with students from other training agencies and a revision of the game.[4] In January 2021, the *Go* training agency enacted a pioneering experimentation by introducing AI training to some students because 16 trainers (out of 36 teachers) were quarantined because of a local surge of COVID-19. Because of labor regulations and other complications, those teachers were not suitable for teaching.[5]

Out of the 287 students observed for the experiment, 151 students (92 boys and 59 girls) received education with human trainers throughout the observation window. Meanwhile, 136 students (82 boys and 54 girls) began receiving training from the AI teacher after January 2021 until the end of the sample period. It is also worth emphasizing that the student–teacher pairs remained unchanged in the experiment except those whose teachers who got quarantined received AI training afterward.

As the disease spread randomly, the event left us with a natural experiment setup.[6] After notifying this event, we collaborated with the *Go* training agency and recovered the training data for two semesters around the shock.[7] Specifically, the first training semester spans from September 2020 to January 2021 and the second from March 2021 to July 2021. There is no training during the Chinese New Year holidays in February 2021. As the shock happened in January 2021, we have four months of observations before the event and five months after the event.

3.1. Treatments

3.1.1. The Control Group (92 Boys, 59 Girls). The natural control of the experiment is the scenario in which human teachers train students throughout the experimental phase.

3.1.2. The Treated Group (82 Boys, 54 Girls). The treatment of the experiment is the scenario in which students are trained by human teachers first and then guided by AI.

It is important to highlight that the assignment of students into control and treated groups is determined by the quarantine statuses of teachers rather than those of students. In addition, the matching protocols for tournaments were the same for the control and treated groups before and after the intervention. Therefore, we do not expect any systematic differences in students' quarantine probability or opponents' characteristics between these groups.

3.2. Hypotheses

We have identified five distinctions between human and AI teaching, which can potentially impact teaching effectiveness. These distinctions encompass teachers' game analysis ability, relevant statistics provision, appearances, interactive features, and emotions. Building on these distinctions, we put forth the first set of hypotheses concerning how these factors might influence teaching effectiveness.

In the domain of strategic board game training, one key to skill improvement lies in the analysis of tournament data (Gobet and Jansen 2006). Human teachers typically base their game analysis on their experience, often reviewing the entire game to identify errors. In contrast, AI teachers process information with greater precision and efficiency (Silver et al. 2016). Therefore, it is plausible that AI's superiority in information processing drives the learning outcomes.

Hypothesis 1.1. *Proficiency in information processing enhances learning outcomes. AI may hold an advantage over human instructors in this aspect.*

Another critical factor is teachers' ability to offer insights on move quality and provide suggestions for improvement (Gobet and Jansen 2006). AI's capability to provide statistics on changes in the expected winning probability and stone count for errors could accelerate students' progress.

Hypothesis 1.2. *Provision of more pertinent statistics improves learning outcomes. AI may have an edge over human teachers in this regard.*

Furthermore, the appearances of human and AI teachers differ. School children may find cartoon-like characteristics more appealing than those of human teachers. Research suggests that visual appeal enhances learning outcomes in various contexts (Cheung and Slavin 2013, Hillmayr et al. 2020). Therefore, we anticipate that the appearance of AI might be more engaging for students and thereby benefit their learning.

Hypothesis 1.3. *A more attractive appearance contributes to better learning outcomes. AI may have an advantage over human instructors in this aspect.*

AI teaching also offers a more interactive visual display compared with human instructors. For instance, the AI teacher highlights errors and suggested moves, also displaying changes in winning probability and stone counts. Existing literature demonstrates that interactive features can improve learning in various contexts (Bai et al. 2020).

Hypothesis 1.4. *Interactive features contribute to better learning outcomes. AI may have an advantage over human instructors in this regard.*

The emotional disposition of a teacher is another pivotal element influencing learning outcomes (Lauermann and Butler 2021). Research shows that teachers' emotions can significantly impact students' confidence, nervousness, attention, willingness to heed advice, and interest in learning (Frenzel et al. 2021). Specifically, teachers' positive and negative emotions may transmit to students through emotional contagion (Hatfield et al. 1993). Consequently, the relative effectiveness of human and AI teaching may depend on the emotional dynamics within the classroom. Nevertheless, comparing the teaching efficacy of human and AI instructors through this emotional lens poses challenges as human teachers might exhibit a broader spectrum of both positive and negative emotions compared with AI teachers.

Hypothesis 1.5. *Positive teacher emotions enhance learning, whereas negative teacher emotions hinder it.*

Based on the aforementioned discussions, we posit the following.

Hypothesis 1. *AI outperforms human teachers because of its advantages in information processing and delivery as well as its more attractive appearance and interactive visual display.*

The second hypothesis delves into the potential gender disparities in response to AI teaching. The emotional dispositions of human and AI teachers are likely to affect boys and girls differently. The AI teacher does not know students' genders; therefore, it can avoid discriminatory emotions that are pervasive among human teachers—one culprit of gender disparity (Alan et al. 2018, Carlana 2019). Thus, the AI may lead to a gender-neutral learning outcome.

Hypothesis 2. *Human teachers may show discriminatory emotions against girls, which leads girls to progress slower than boys. The AI teacher improves girls' relative Go performance by showing nondiscriminatory emotions.*

Boys and girls may also differ in the responsiveness of other human–AI differences mentioned in Hypotheses 1.1–1.4. We examine these possibilities with a survey.

3.3. Survey on Teaching-Related Questions

To understand the channels that may drive the differences in the learning outcomes between AI and human teaching, we surveyed 86 students enrolled in the *Go* training agency in the spring semester of 2023 who had experienced the same AI training as the treated group in the experiment. The survey was conducted using Wenjuanxin, a Chinese equivalent of Qualtrics, from April to May 2023, and we asked students questions regarding their experience with human and AI teaching. Survey participants finished their answers on the computers from the *Go* training agency in their revision classes. We informed students that their answers would be anonymous and no one from the *Go* agency could access individual responses. Survey questions are listed in Online Appendix F.1.

4. Data and Variable Construction

We collect three sets of information from the experiment: student and teacher attributes, students' *Go* tournament data, and video recordings for revision classes. The following sections introduce the definition of these variables in detail.

4.1. Student and Teacher Data

We obtain demographic data from all 287 students and their 36 teachers, collecting data on students' age, gender, years of study (number of years they learned from the agency), and *dan* level. We also retrieve teachers' age, gender, and experience (how many years they have been teaching *Go*).

We show the summary statistics of student and teacher characteristics for the entire sample and two subsamples (the treated and control groups) in panels A and B of Table 1, respectively. On average, students are 10.22 years old with 0.93 years studying in *Go* from the agency and a *dan* rank of 1.26. Among all students, about 61% are boys. As for teachers, they are on average 22.06 years old with 1.89 years of experience in *Go* training, and 56% of them are men. There is no significant difference in students' and teachers' characteristics between treated and control groups.

4.2. Tournament Data

As mentioned earlier, every week's first part of training is a tournament with one of the peers. We collect students' complete playing records during the sample period, including detailed information on each game they played, such as the sequence of moves and the binary game outcome (win/lose). The binary win/lose variable is a proxy for students' *Go* skills, but it is noisy because the outcome of games may depend on opponents' performance during the game, and it does not capture the degree of win/lose. Moreover, analyzing

Table 1. Summary Statistics

	Full sample		Control (C)		Treated (T)		
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation	C − T
Panel A. Student data							
Age	10.22	2.14	10.24	2.19	10.20	2.10	0.04
Study, years	0.93	1.11	0.93	1.12	0.92	1.10	0.01
dan (1–5)	1.26	1.18	1.21	1.25	1.32	1.11	−0.10
Boy	0.61	0.49	0.61	0.49	0.60	0.49	0.01
Panel B. Teacher data							
Age	22.06	3.83	21.50	3.98	22.75	3.64	−1.25
Experience, years	1.89	0.78	1.75	0.72	2.06	0.85	−0.31
Male	0.56	0.50	0.60	0.50	0.50	0.52	0.10
Panel C. Game data							
Quality	5.49	3.25	5.40	3.45	5.58	3.02	−0.18***
Quality (opening)	3.58	1.86	3.52	1.97	3.65	1.72	−0.12***
Quality (middle)	4.34	2.33	4.27	2.47	4.42	2.15	−0.15***
Quality (ending)	8.33	5.63	8.15	5.95	8.53	5.24	−0.38***
Number of errors	19.47	4.90	19.60	5.05	19.32	4.72	0.28***
Number of critical errors	2.32	0.64	2.34	0.67	2.30	0.60	0.04***
Magnitude of errors	10.39	2.22	10.43	2.31	10.34	2.11	0.09***
Win	0.51	0.50	0.50	0.50	0.52	0.50	−0.02***
White (late mover)	0.50	0.50	0.51	0.50	0.50	0.50	0.01
Panel D. Video data							
Visual-positive	0.16	0.12	0.16	0.13	0.15	0.10	0.02***
Visual-negative	0.12	0.10	0.13	0.11	0.10	0.08	0.02***
Vocal-positive	0.10	0.09	0.11	0.10	0.09	0.07	0.02***
Vocal-negative	0.01	0.01	0.01	0.01	0.01	0.01	0.00***
Verbal	4.80	1.50	4.95	1.51	4.64	1.47	0.31***

Notes. This table provides the mean and standard deviation of each variable discussed in this paper for the whole population and treatment subsamples. We report information regarding students in panel A, teachers in panel B, students' moves in tournaments in panel C, and video recording of revision sessions in panel D. See the main text for more details on variable construction and definition.

***, **, and * indicate statistical significance for the *t*-test testing the null hypothesis that the control and treated groups have the same average value at the 1%, 5%, and 10% levels, respectively.

dummy dependent variables with probability models (e.g., logit regression) can be computationally demanding and may yield unstable estimates, particularly with a large sample size. Therefore, analyzing nonbinary variables strengthens the results and offers additional insights. In the following section, we introduce three sets of variables summarizing information of all moves in each game to complement the binary game outcome.

4.2.1. Move Quality. We use the KataGo machine learning algorithm to evaluate the quality of moves players made in each game in our data set.[8] KataGo, initiated by David J. Wu, is one of the most potent open-source *Go* bots available online (Wu 2019). Because of its advantage over most other AI, KataGo is widely used by amateur and professional players.

KataGo can evaluate the change in the probability of winning and the final stone count for all alternative moves at any point during a game, enabling us to compare the difference in winning probability (stone count) between a player's move and KataGo's suggested move. Based on these comparisons, KataGo helps us to construct three sets of variables to measure the quality of moves.

The first set of variables considers the average quality of a group of moves made by a player in one game. KataGo constructs the variable *Average Move Quality* to measure the average quality of moves in the whole game as well as analogous measures for the average move quality at different game stages, including the opening, middle, and ending of a game relative to professionals' performance. The variable ranges from 0 (i.e., beat 0% of professionals) to 100 (i.e., beat 100% of professionals). Without further specification, the variable hereafter denotes the average move quality of all moves a player makes in a game.

The second set of variables counts the number of errors/critical errors a player makes in a group of moves. KataGo classifies all moves into different categories based on the change in the probability of winning and the stone count. We construct the variable *Number of Errors* to count the number of moves that reduce the winning possibility of the player by 10% to 20% and the variable *Number of Critical Errors* to measure the number of moves that drop the player's winning probability by more than 20%.

The third set of variables addresses the magnitude of all errors (including critical errors). KataGo computes the expected final stone count difference between a player's move and its recommended move for each move a player makes. We take the average value of these numbers across the game and construct the variables *Magnitude of Errors* to measure the size of reduction in the final stone count from students' suboptimal moves.[9]

4.2.2. Tournament Data Summary. Panel C of Table 1 provides summary statistics on variables in the tournament data with a total of 22,382 student–tournament observations. The average move quality is 5.49, whereas the quality at the opening, middle, and ending, on average, is 3.58, 4.34, and 8.33, respectively. The average number of errors and critical errors are 19.47 and 2.32, and the average magnitude of errors is 10.39. The winning probability is about 51%, and the probability of a player taking a white stone is 50%.

The treated and control groups have substantial differences in game performance. Compared with the control group, the treated group has better average move quality and winning probability, making fewer errors at all game stages. As the comparisons here are for all matches (i.e., before and after the AI intervention), we need to compare the group differences before and after the intervention to learn the effectiveness of AI teaching.

4.3. Video Data

As mentioned, the second part of training every week is replaying the tournament just played. The teacher evaluates the student's moves in a one-on-one video meeting, which the training agency records. We collect the training videos of both human and AI simulated teachers. After dropping videos unable to analyze, our final sample consists of 20,279 video records.[10] We extract information from these records to construct interpretable variables with machine learning algorithms.

4.3.1. Processing Video Data. Following the literature in communication and psychology, we decompose videos into a three-dimensional structure, including visual, vocal, and verbal information, and represent them in digital format (Krauss et al. 1981, Wallbott and Scherer 1986). We next employ machine learning algorithms to analyze these pieces of information and construct measures for teachers' emotions revealed in online classrooms.

For the visual dimension, we adopt the face detection algorithms provided by the Megvii platform to identify facial emotions embedded in each video. The Megvii algorithm detects facial landmarks and movements and measures emotions in six aspects: happiness, sadness, anger, fear, disgust, and contempt. To capture positive visual emotions, we create *Visual-Positive* measuring the proportion of time the teacher's facial emotion is positive (happiness). To capture negative emotions, we take the total proportion of times for sadness, anger, fear, disgust, and contempt as a composite facial emotion and create *Visual-Negative* measuring the degree of facial emotion being negative (sadness, anger, fear, disgust, and contempt).[11]

For the vocal dimension, we extract the video audio streams and treat them as time series of amplitudes. We focus on the intonation of words and sentences using the speech recognition algorithm developed by Giannakopoulos (2015).[12] The algorithm approximates teachers' moods with the rise and fall in their tone along three dimensions: positive, negative, and neutral. We compute the proportion of time in the video that a teacher shows certain vocal emotion and create two variables, *Vocal-Positive* and *Vocal-Negative*, that measure the degree to which vocal emotion is positive or negative.

For the verbal dimension, we use the speech-to-text algorithm developed by Baidu AI to convert teachers' speech into verbal context.[13] The converted verbal contents include a list of words, time stamps of these words, and punctuation. For reproducibility and transparency, we measure the teachers' sentiment using the machine-trained Hedonometer semantic analysis dictionary on the verbal corpus (Dodds et al. 2011, Kloumann et al. 2012).[14] This dictionary is commonly used in emotional analysis and includes a comprehensive Chinese word list that helps us to measure the emotion reflected by the texts (Dodds et al. 2015). It contains 10,014 words and assigns a happiness score ranging from one to nine with one indicating a strong negative and nine a strong positive emotion for each word. We construct the *Verbal-Emotion* variable as the sum of happiness score by all words scaled by the total number of words in each video.[15] Our results are robust when we use the natural language processing algorithm provided by Tencent that performs sentiment analysis based on computational linguistics techniques.

4.3.2. Video Data Summary. Panel D of Table 1 presents the summary statistics for variables related to training videos. On average, the values for *Visual-Positive* and *Visual-Negative* are 0.16 and 0.12 for the full sample, indicating that, during the sample period, teachers show positive and negative visual features for 16% and 12% in replaying classes. The mean values for *Vocal-Positive* and *Vocal-Negative* are 0.10 and 0.01, showing that the teachers show positive and negative vocal features for 10% and 1% when replaying tournaments with students. The average value for *Verbal-Emotion* is 4.80 for the full sample, consistent with the average emotion of Chinese urbanites' expressions on social media (Zheng et al. 2019).

We then compare these video features between the treated and control groups. Statistically significant differences are observed for all emotional variables. The AI trainer exhibits a more neutral emotional status compared with human teachers with a smaller proportion of positive and negative emotions observed in the treated group's training video compared with the control group.

4.4. Survey on Students' Perceptions of the AI Teacher

Whereas our facial, vocal, and verbal analyzing algorithms have been established to effectively evaluate human emotions, their ability to analyze the emotions of the cartoon animated AI teacher remains uncertain. To validate the use of these algorithms in studying AI's emotions, we conducted a survey involving 57 students enrolled in the *Go* training agency from April to May 2023. During the survey, the students watched seven video clips of AI teaching and were requested to document their perceptions of the AI teacher's visual, vocal, and verbal emotions for each clip on paper. Our findings reveal a statistically significant correlation between the students' responses and the scores generated by the algorithms. Survey questions are listed in Online Appendix F.2.

Additionally, we examined the students' perception of the AI teacher's gender by asking them to choose among female, male, and gender-neutral based on its appearance and sound. The majority of students indicated a gender-neutral perception for both aspects with a balanced percentage of students perceiving the character as male or female. For more detailed survey results, please refer to Online Appendix C.4.

5. Empirical Methodology

We study the effects of the AI training using the difference-in-differences (DID) model, in which we compare students' performance in treated and control groups before and after the experimental intervention. The role of the control group is to establish a counterfactual of what would have happened to students' performance in the treated group if they had received training from human teachers throughout the observation window.[16]

As discussed earlier, we observed 287 students' training records over 10 months, creating a panel data set with 22,382 student–game observations. Given the data structure, we specify the baseline econometric model at the student–game level and perform the following econometric specification:

$$y_{ijt} = \alpha + \beta \text{Treated}_i \times \text{After}_t + \gamma_i + \delta_t + \epsilon_{it}, \quad (1)$$

where y_{ijt} is a series of game-level dependent variables, including average move quality, the number of errors/critical errors, the magnitude of errors, and game outcome for player i in month t at game j; Treated_i is a dummy variable that equals one if the player i received the AI training, Treated_i equals one if month t is after the start of the experimental intervention (i.e., from March to July 2021) and zero otherwise. Our specification exploits the panel data structure and includes a full set of fixed effects to alleviate the endogeneity concerns. Specifically, we add the player fixed effects γ_i to control for the observed and unobserved player-specific variables and month fixed effects δ_t to absorb time-specific confounds that may affect dependent variables. We also control for class and teacher fixed effects to take care of confounding factors specific to classes and teachers.[17]

When the dependent variable y_{ijt} is a dummy variable (i.e., win/lose), we use the logit model with individual and time effects to estimate the treatment effect. As shown in Neyman and Scott (1948), fixed effects panel data methods in logit regressions can be severely biased because of the incidental parameter problem. We use the panel jackknife bias correction method to tackle these problems (Dhaene and Jochmans 2015).

To study potential heterogeneous effects across genders, we further consider the triple difference extension of Equation (1) that captures gender differences in responsiveness to AI training. Specifically, we include all interaction terms involving Boy_i, Treated_i, and After_t in the analysis:

$$\begin{aligned} y_{ijt} = \alpha &+ \beta_1 \text{Treated}_i \times \text{After}_t + \beta_2 \text{Boy}_i \times \text{After}_t \\ &+ \beta_3 \text{Treated}_i \times \text{After}_t \times \text{Boy}_i + \gamma_i + \delta_t + \epsilon_{it}, \end{aligned} \quad (2)$$

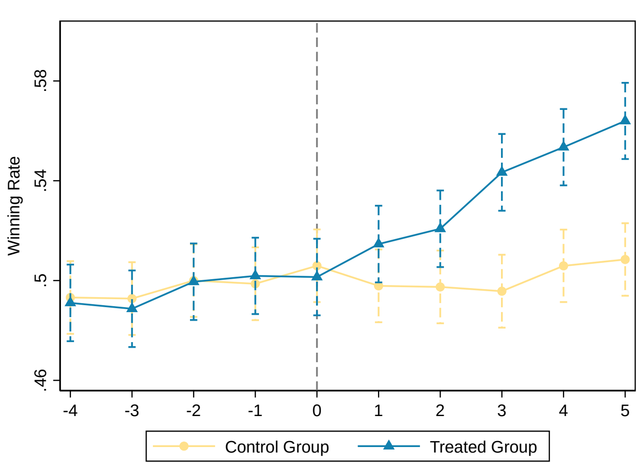
where Boy_i is a dummy variable that equals one if the student is a boy and the definitions of other variables are identical to that in Equation (1). The parameter β_3 is the triple difference estimator, which quantifies the differential treatment effects of AI training between boys and girls.

6. Empirical Results

6.1. Effectiveness of AI Training

6.1.1. Winning Probability. We first plot the unconditional winning rate of the control and treated groups in Figure 1 to examine the effectiveness of AI training. Before the experimental intervention started, we find both control and treated groups performed similarly; the winning probability of both groups was around 50%. After the intervention, the winning chance of the treated group increased steadily from 50% to 56% after five months of AI training, but the winning rate of the control group remained at 50% with no clear trend.

Figure 1. (Color online) Winning Ratio



Notes. This figure shows the winning rates (raw rates) of the control and treated groups before and after the experimental intervention. The vertical axis plots the winning probability, and the horizontal axis shows the time relative to the start of the intervention. The dashed error bars depict the 95% confidence interval.

Although there was no observed change in the winning probability of the control group, this does not necessarily imply that their *Go* skills remained stagnant throughout the observation period. The winning probability merely provides insight into their relative *Go* performance rather than an absolute measure of their abilities. To gain a more accurate understanding of the students' objective skill levels, we analyze move quality metrics in the subsequent section.

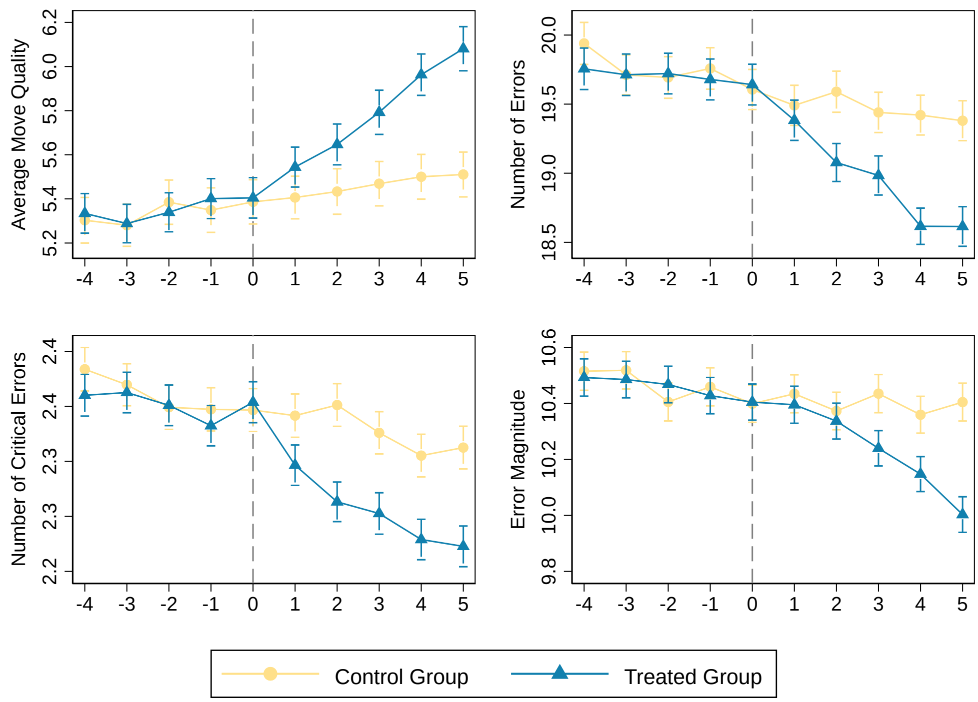
6.1.2. Move Quality. We examine variables that serve as proxies for students' move quality before and after the treatment intervention in both the control and treated groups. Specifically, we utilize four measures to assess players' absolute performance. The first measure, average move quality, provides an overall assessment relative to professional players. The remaining three measures—number of errors, number of critical errors, and error magnitude—capture the quantity and severity of errors. In Figure 2, we present the evolution of these variables, analogous to Figure 1. The panel depicting average move quality reveals a slight and steady upward trend in the control group's performance over the observation period, suggesting students improve gradually without AI. The treated group's average move quality initially mirrored that of the control group. However, following the introduction of AI training, the treated group experienced a sharp increase in this measure. Just two months after the intervention, the treated group's score surpassed that of the control group significantly, and the gap continued to widen.

We then examine the error measurements in tournaments. In the control group, errors decreased gradually over time as students learned to avoid mistakes with human teachers. Before AI training, there were no significant differences between the treated and control groups in these measures. However, after the introduction of AI, the treated group showed a more pronounced reduction in errors, indicating a significantly lower likelihood of making inferior moves compared with the control group.

Figures 1 and 2 suggest that AI training outperforms human teachers: the AI teacher helps students to improve the winning probability by improving their move quality and reducing errors. To quantify the treatment effect size and test the robustness of these results, we perform the regression analyses shown in Equation (1) and present the results in Table 2.

Column (1) shows the average treatment effect of AI training on the tournament binary outcome (win/lose). As mentioned in Section 5, we use the logit fixed effect estimator with jackknife bias corrections to estimate the model and only plot the average treatment effect in the table. AI training improved the winning probability by 3.5 percentage points ($p < 0.01$), a 7% improvement on a base of 50% for the control group (see panel C in

Figure 2. (Color online) Measures of Performance



Notes. This figure plots four measures of the control and treated groups' performances (raw rates) before and after the start of the experimental intervention. These four measures are average move quality, number of errors, number of critical errors, and error magnitude. The vertical axis plots the values of each measure, and the horizontal axis shows the time relative to the start of the intervention. The error bars depict the 95% confidence interval.

Table 1). Column (2) presents the average treatment effect of AI training on average move quality. The point estimate is 0.328 and statistically significant at the 1% level.[18] In columns (3) and (4), we focus on number of errors and critical errors. The corresponding coefficient estimates indicate that, over the AI training phase, the number of errors and critical errors of the treated group further dropped by 0.490 and 0.067 times, respectively, and both estimates are statistically distinguishable from zero. We then look at the treatment effect on the magnitude of errors in column (5) and find the point estimate is −0.173 and statistically significant at the 1% level. These estimation results confirm the advantage of AI training in improving students' performance as the coefficients

Table 2. Estimation of the Average Treatment Effect

	Dependent variable				
	Game outcome (1)	*Average move quality* (2)	*Number of errors* (3)	*Number of critical errors* (4)	*Magnitude of errors* (5)
After × *Treated*	0.035***	0.328***	−0.490***	−0.067***	−0.173***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Constant	—	5.409***	19.584***	2.340***	10.429***
	—	(0.00)	(0.00)	(0.00)	(0.00)
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes
Teacher fixed effects	Yes	Yes	Yes	Yes	Yes
Student fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	22,382	22,382	22,382	22,382	22,382

Notes. This table shows the estimation results of the effectiveness of AI training in improving students' performance. The dependent variables are *Game Outcome* (dummy variable for winning), *Average Move Quality*, *Number of Errors*, *Number of Critical Errors*, and *Magnitude of Errors* in columns (1)–(5), respectively. The independent variable *After* is a dummy variable specifying whether the game took place after the introduction of AI training, and *Treated* is a dummy variable determining whether the student is in the treated group. Standard errors are clustered at the student level. *p*-values are reported in parentheses.

***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

attached to the DID estimator (Treated × After) are significantly positive for the winning probability and significantly negative for error measures.

6.1.3. Further Analyses. The preceding regression analyses provide an overview of the average treatment effects. However, it remains unclear how these effects evolve over time. To shed light on the performance differences between the treated and control groups over time, we employ a generalized DID specification of Equation (1). We interact the model with dummy variables representing different months and compare the performance differences between the treated and control groups chronologically. We designate the month prior to the intervention (i.e., $t = 0$) as the baseline time, and subsequent monthly comparisons are relative to the group difference at $t = 0$.

We plot the estimated coefficients over time in Figure 3.[19] The data suggests no preexisting performance gap between the control and treated groups as the coefficients of DID estimators for months before the experimental intervention are not significantly different from zero. However, after the introduction of the AI training, the treated group performed better than the control group, and the gap between them expanded persistently. This is consistent with the patterns in previous figures.

We also perform other heterogeneous analyses to study whether a particular group of students drives the main results. We decompose all students into different groups categorized by age, study duration, and *dan* rank and perform analogous analyses to these subsamples. From estimation results in Online Appendix C.3, we find the results above hold qualitatively across these groups, indicating that the effects of AI training on improving performance applies to students with various characteristics.[20]

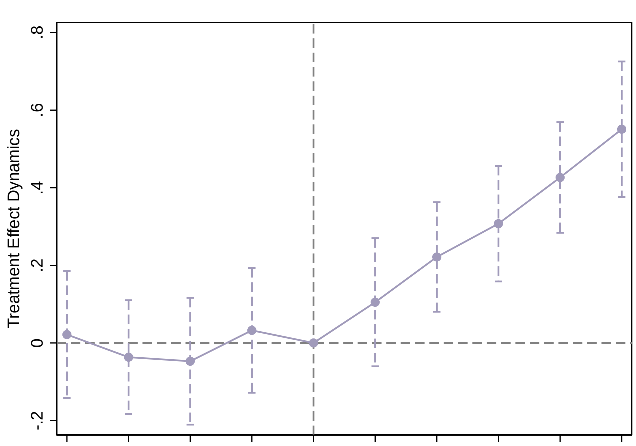
Result 1. The AI outperforms human teachers in improving students' performance.

6.2. AI Training and Gender Performance Gap

Now that we find the merit of AI training in improving students' *Go* performance, it is still unclear whether and to what extent AI affects the gender performance gap in playing. To explore this question, we replicate the analyses in Section 6.1 by the gender of players and plot Figure 4 to study how boys and girls respond to AI differently. In the control group, boys consistently have a higher winning rate than girls, indicating the existence of gender disparity in *Go* training. Both boys' and girls' winning probabilities have no clear time trend along the observation window and fluctuate around their respective means. The winning probability of the treated group is similar to that of the control group before AI training. However, after the intervention, boys and girls in the treated group improved their winning probability over time and became significantly better than their counterparts in the control group. It is also worth noting that girls' winning probability improved faster than boys', diminishing the gender performance gap five months after the intervention.

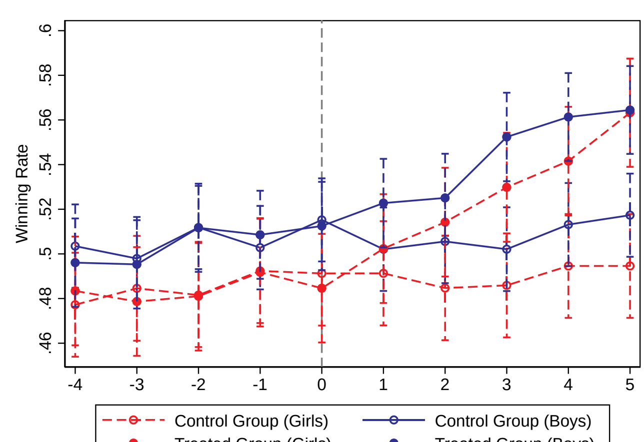
We then focus on move quality measures in Figure 5. From the average move quality panel, we find boys and girls in the control group had a modest upward trend in their average move quality, and the move quality gap between them is steady over time. The panel also suggests that, in the treated group, both boys and girls have a quicker move quality improvement after receiving the AI training. However, girls' rate of increase is steeper

Figure 3. (Color online) Evolution of Treatment Effects



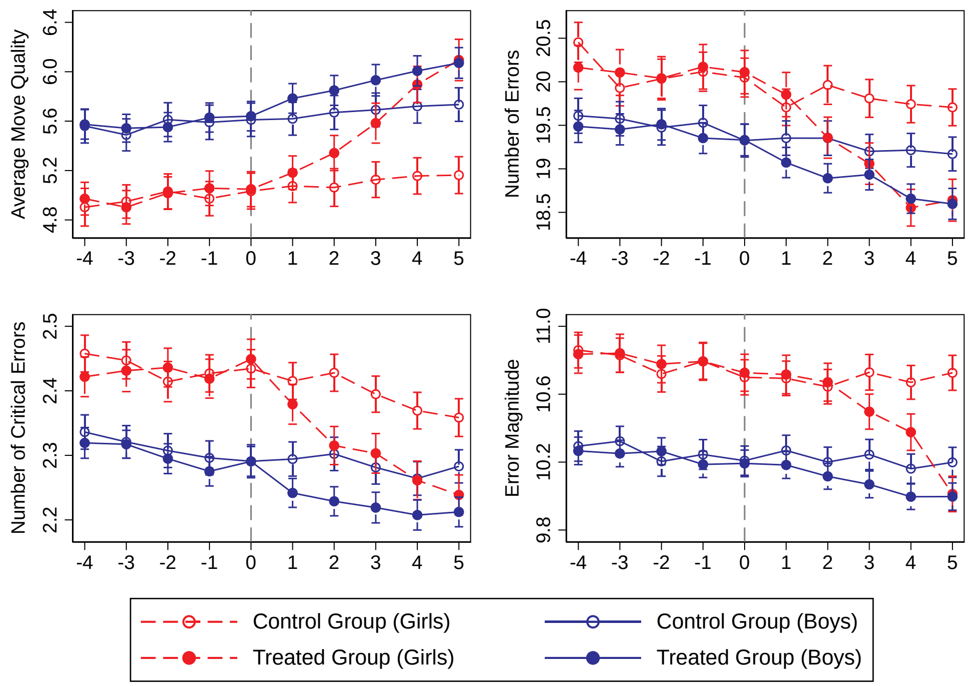
Notes. This figure shows the estimation coefficients for the dynamic difference-in-differences regressions that explore the impact on average move quality between the treated and control groups over the experimental window. The vertical axis plots the estimation coefficients on average move quality in each month of the observation window, and the horizontal axis shows the time relative to the start of the intervention. The error bars depict the 95% confidence interval.

Figure 4. (Color online) Winning Ratio by Gender



Notes. This figure shows the winning rates (raw rates) of the control and treated groups before and after the experimental intervention by gender. The vertical axis plots the winning probability, and the horizontal axis shows the time relative to the start of the intervention. The dashed error bars depict the 95% confidence interval.

Figure 5. (Color online) Measures of Performance by Gender



Notes. This figure plots four measures of the control and treated groups' performances (raw rates) before and after the start of experimental intervention by gender. These four measures are average move quality, number of errors, number of critical errors, and error magnitude. The vertical axis plots the values of each measure, and the horizontal axis shows the time relative to the start of the intervention. The error bars depict the 95% confidence interval.

than boys', eliminating the preexisting gender gap just five months after the intervention. The data also suggests that, at the end of the observation window, girls receiving AI training outperformed boys in the control group.

The evolution of three error measures follows an analogous pattern. For the control group, both boys' and girls' error incidences and magnitudes reduced gradually over the experimental window, and boys consistently outperformed girls. For the treated group, the patterns for boys and girls were similar to that of the control group before the intervention. However, after the introduction of AI, both boys and girls had a sharper reduction in the error measures with girls reducing errors at a faster rate than boys. After five months of AI training, the preexisting gender gaps in error making have been closed.

We then perform regression analyses quantifying to what extent that AI training can reduce the gender performance gap as in Equation (2). In Table 3, we find that the coefficient attached to the triple difference estimator ($\text{Treated}_i \times \text{After}_t \times \text{Boy}_i$) is consistent with what we observe from the figures: AI training can reduce the preexisting gender performance gap. Specifically, we show the differential effects of AI training on game outcome for boys and girls in column (1). The coefficient on the triple difference estimator is -0.059 ($p < 0.01$), which implies that the average increase in winning probability from AI teaching is larger for girls than boys. In column (2), we find boys' average move quality improved less than girls with a coefficient on the triple difference estimator of -0.250 ($p < 0.01$). Similarly, in columns (3)–(5), the data suggests that girls reduced the incidence and magnitude of errors and critical errors more significantly than boys after AI training.

We also perform dynamic analyses to study how the introduction of AI benefits boys and girls differently. We estimate two separate regressions by the gender of students and plot the coefficients in each month for each gender in Figure 6. We find boys and girls in the treated group achieved similar learning outcomes as those in the control group before AI training, respectively, as the DID coefficients are insignificant for months before $t = 0$. However, after the introduction of the AI teacher, both genders in the treated group learned quicker than the control group. After $t = 2$, the control–treatment difference became significant for both boys and girls. It is also conspicuous that girls' performance improved faster than boys with the help of AI training; the treatment effect for girls is significantly larger than for boys after four months.[21]

Result 2. The AI training ameliorates the preexisting gender inequality.

6.3. Robustness Checks

6.3.1. Opponents' Characteristics. Estimation results in Tables 2 and 3 may depend on the opponents'

Table 3. Estimation of the Gender Difference in the Treatment Effect

	Dependent variable				
	Game outcome (1)	*Average move quality* (2)	*Number of errors* (3)	*Number of critical errors* (4)	*Magnitude of errors* (5)
After × *Treated*	0.048**	0.479***	−0.691***	−0.088***	−0.251***
	(0.02)	(0.00)	(0.00)	(0.00)	(0.00)
After × *Boy*	−0.029	−0.032	0.097	0.019	0.051
	(0.28)	(0.58)	(0.32)	(0.25)	(0.26)
After × *Treated* × *Boy*	−0.059***	−0.250***	0.336**	0.036	0.131**
	(0.00)	(0.01)	(0.02)	(0.15)	(0.03)
Constant	—	5.418***	19.555***	2.334***	10.413***
	—	(0.00)	(0.00)	(0.00)	(0.00)
Month fixed effects	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes
Teacher fixed effects	Yes	Yes	Yes	Yes	Yes
Student fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	22,382	22,382	22,382	22,382	22,382

Notes. This table estimates to what extent AI training affects boys and girls differently. The dependent variables are *Game Outcome* (dummy variable for winning), *Average Move Quality*, *Number of Errors*, *Number of Critical Errors*, and *Magnitude of Errors* in columns (1)–(5), respectively. The independent variable *After* is a dummy variable specifying whether the game took place after the introduction of AI training, *Treated* is a dummy variable determining whether the student is in the treated group, and *Boy* is a dummy variable on students' gender. Standard errors are clustered at the student level. p-values are reported in parentheses.
***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

characteristics. In a set of robustness analyses, we control for observed opponents' characteristics, and Results 1 and 2 remain unchanged. Additionally, we find that neither boys nor girls are significantly impacted by their opponents' genders. Interested readers may find more discussion in Online Appendix A.4.

6.3.2. Multiple Hypotheses Testing. The regression analyses in Tables 2 and 3 may suffer from the multiple hypothesis testing criticism because we include multiple outcomes (game outcome, average move quality, number of errors, number of critical errors, and magnitude of errors). To address this issue, we adopt four approaches (Westfall and Young 1993, Anderson 2008, Romano and Wolf 2016, Barsbai et al. 2020) to correct for multiple hypothesis testing problems and find that the results remain robust. Interested readers can find detailed information in Online Appendix A.5.

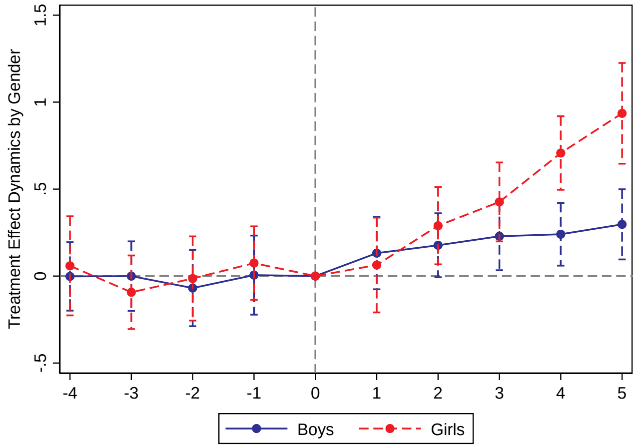
7. Potential Channels

We find AI's merit in improving students' *Go* performance, particularly for girls. However, the mechanisms through which AI achieves this outcome remain unclear. In this section, we investigate potential explanations as in the hypothesis section by conducting surveys and additional analyses.

7.1. Information Processing

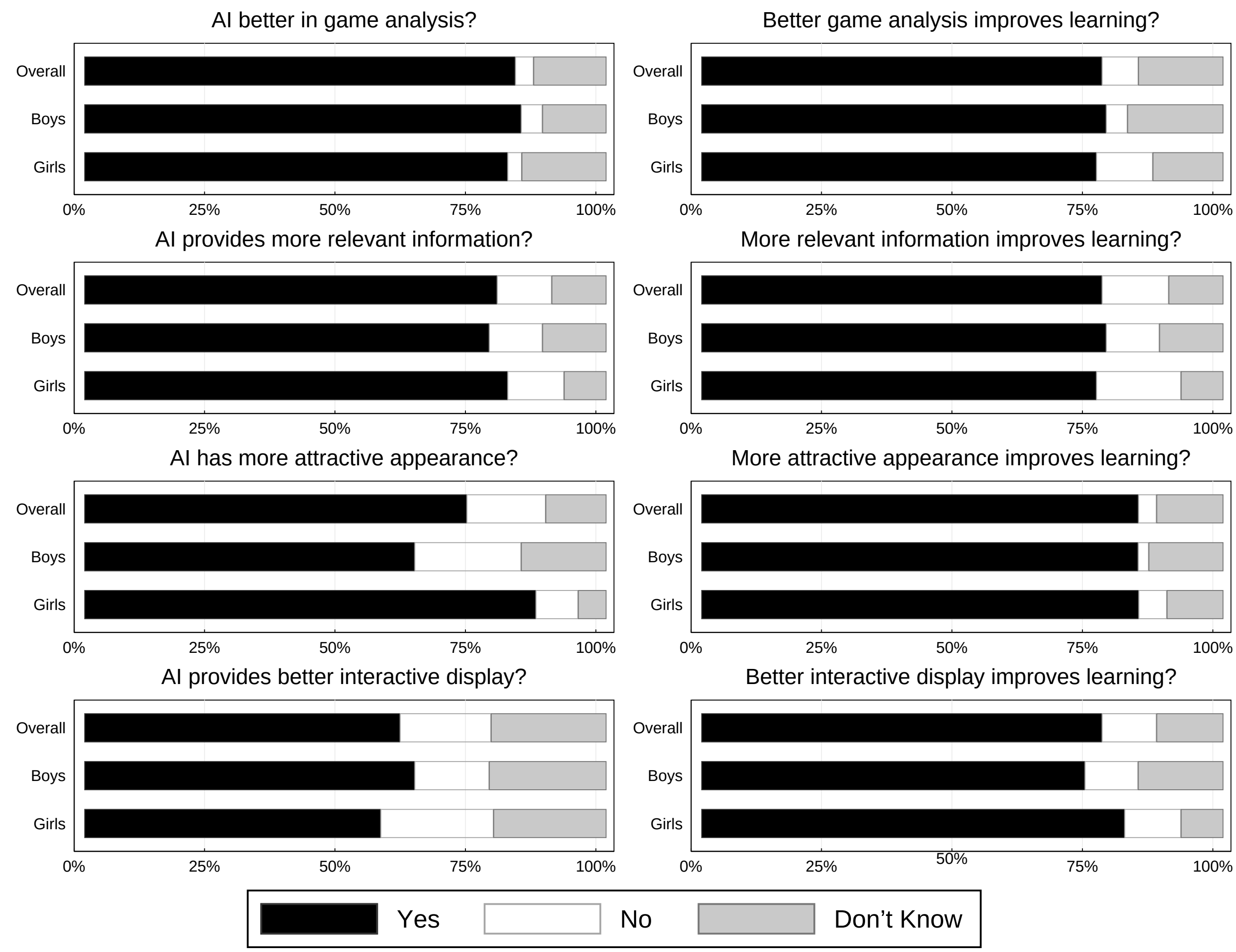
As discussed in Hypothesis 1.1, AI has a clear advantage in analyzing the game over human teachers. To explore whether better game data analysis translates into improved teaching outcomes, we asked students whether they believed the AI teacher could analyze the game data more comprehensively than human teachers. The left panel in the first row of Figure 7 plots the results and suggests 82.6% of respondents responded "yes" to this question; 83.7% of boys and 81.1% of girls responded positively with no significant gender differences in the rates (p = 0.75).[22] We then

Figure 6. (Color online) Evolution of Treatment Effects by Gender



Notes. This figure shows the estimation coefficients for dynamic difference-in-differences regressions that explore the impact on average move quality between the treated and control groups over the experimental window by gender. The vertical axis plots the estimation coefficients on average move quality in each month of the observation window, and the horizontal axis shows the time relative to the start of the intervention. The error bars depict the 95% confidence interval.

Figure 7. Survey Results on Human–AI Distinctions



Notes. This figure shows survey results of students' perception of AI's potential advantages in game analysis, information provision, appearance, and interactive features and their belief on whether these features can improve the learning outcome. The detailed survey questions are listed as Q5 to Q13 from Online Appendix F.1.

asked students whether they believed teachers' ability to analyze game data positively affects their learning outcomes. The right panel in the first row of Figure 7 presents the results. In sum, 76.7% of respondents agreed with the statement, whereas the numbers are 77.6% and 75.7% for boys and girls ($p = 0.84$), respectively.

Result 1.1. Students believe the AI teacher has better game-analyzing ability than human teachers, which helps them improve faster.

7.2. Information Provision

Per Hypothesis 1.2, AI also outperforms human teachers in providing students with outcome-related statistics. We asked students whether the AI provides more relevant statistics and information than human teachers. The left panel in the second row of Figure 7 plots the results. Overall, 79.1% of respondents answered "yes" to this question. There were no significant gender differences in the rates ($p = 0.69$) with 77.6% of boys and 81.1% of girls responding positively. We also asked students whether providing these statistics improves their learning outcomes. The right panel in the second row of Figure 7 plots the results. Overall, 76.7% of respondents agreed with the statement, and the numbers are 77.6% and 75.7% for boys and girls ($p = 0.84$), respectively.

Result 1.2. Students suggest that the AI teacher provides more relevant statistics than human teachers, which improves learning outcomes.

7.3. Teacher Appearance

Echoing Hypothesis 1.3, AI's cartoon appearance may be more attractive to students than human teachers. We asked students to compare the attractiveness of appearance between the AI and human teachers in the survey. The left panel in the third row of Figure 7 plots the results: 73.3% of respondents believe the cartoon figure has a more attractive appearance; 63.3% of boys and 86.5% of girls responded positively to this question, and the gender difference is significant ($p = 0.02$). Next, we asked whether teachers with more attractive appearances improve learning and plotted the results on the right panel in the third row of Figure 7: 83.7% of students are affirmative to the question, and rates are 83.7% and 83.8% for boys and girls ($p = 0.99$), respectively.

Result 1.3. Students believe that the AI's cartoon appearance is more attractive than human teachers, and more attractive appearances improve learning. Girls have a stronger preference for the AI's appearance than boys; they may benefit more than boys from AI teaching through this channel.

7.4. Interactive Features

Responding to Hypothesis 1.4, we asked survey participants whether the AI offers more interactive features, such as highlighting errors and suggested moves, than human teachers. The left panel in the fourth row of Figure 7 plots the results. Overall, 60.5% of respondents perceive that AI provides more interactive features than human teachers. Turning the focus to the gender angle, we find no significant gender differences in the responses: 63.3% of boys and 56.8% of girls responded "yes" to the question ($p = 0.54$). We also asked students whether these features improve learning outcomes on the right panel in the fourth row of Figure 7, respectively; 76.7% of students believe interactive features are positively related to learning outcomes, and boys and girls do not differ in the positive response rate (73.5% versus 81.1%, $p = 0.41$).

Result 1.4. Students believe that the AI teacher provides more interactive features than human teachers, which improves learning outcomes.

7.5. Teacher Emotions

7.5.1. Teachers' Emotions and Students' Characteristics. As conjectured in Hypotheses 1.5 and 2, teachers' emotions can affect learning outcomes and the gender performance gap. By convention in the psychological literature (Krauss et al. 1981, Duckworth et al. 2007), we analyze teacher emotions embedded in class recordings' visual, audio, and verbal information and perform regressions to examine if there is a correlation between teachers' emotions and students' gender for both human and AI teachers, controlling for students' performance in previous games. Following the literature on emotions (Durlak et al. 2011, Hu and Ma 2021), we present the standardized coefficients for regressions studying teacher emotions (i.e., Tables 4–6).

First, we focus on columns (1)–(4) in Table 4 to study how teachers' facial emotion correlates with students' characteristics. The visual data find strong evidence that human teachers are significantly more likely to show positive emotions (column (1)) and less likely to show negative emotions (column (3)) to boys and students with higher *dan*. However, the AI teacher does not change emotions with observed student characteristics (columns (2) and (4)). Columns (5)–(8) plot the estimation results based on teachers' vocal information and find a similar pattern; human teachers tend to favor boys and students with higher *dan*, whereas AI's emotion is impartial to students. The same pattern also applies to teachers' verbal emotions as in columns (9) and (10). In sum, human teachers show more positive and less negative emotions toward boys, whereas AI teachers do not show such gender disparity; AI's emotion is independent of students' characteristics.[23]

Table 4. Determinants of Teachers' Emotional Status

	Dependent variable									
	Visual-positive		*Visual-negative*		*Vocal-positive*		*Vocal-negative*		*Verbal*	
Teacher	Human (1)	AI (2)	Human (3)	AI (4)	Human (5)	AI (6)	Human (7)	AI (8)	Human (9)	AI (10)
Boy	0.007**	−0.001	−0.007**	−0.000	0.015***	−0.001	−0.001***	−0.000	0.384***	0.070
	(0.01)	(0.75)	(0.01)	(0.95)	(0.00)	(0.33)	(0.00)	(0.27)	(0.00)	(0.23)
Age	0.000	0.000	−0.000	0.000	0.000	−0.000*	0.000	−0.000	0.002	−0.008
	(0.93)	(0.40)	(0.79)	(0.86)	(0.96)	(0.07)	(0.60)	(0.32)	(0.84)	(0.67)
Study	0.001	0.000	0.000	0.000	0.000	0.001	−0.000	−0.000	0.002	0.009
	(0.62)	(0.99)	(0.88)	(0.94)	(1.00)	(0.12)	(0.11)	(0.54)	(0.90)	(0.79)
dan	0.007	−0.003	−0.012***	−0.000	0.006	−0.003*	−0.001*	−0.000	0.100*	0.054
	(0.15)	(0.37)	(0.00)	(0.84)	(0.14)	(0.09)	(0.06)	(0.68)	(0.08)	(0.28)
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Teacher fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,630	4,812	13,628	4,812	13,628	4,812	13,630	4,812	13,630	4,809

Notes. This table estimates how much students' characteristics affect their human and AI teachers' emotions expressed in class recordings. The dependent variables capture teachers' facial, vocal, and verbal emotions, and the independent variables include students' gender, age, years of study, and *dan* rank. Game controls include three variables capturing the average moving quality for the previous one, three, and seven games, respectively. Standard errors are clustered at the student and teacher levels. *p*-values are reported in parentheses.

***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5. Human Teachers' Emotional Status and Subsequent Student Performances

	Dependent variable							
	Average move quality							
Teacher	Human							
Student	Girls and Boys						Girls	Boys
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Visual-positive	0.009**					0.007	0.003	0.013*
	(0.03)					(0.14)	(0.51)	(0.07)
Visual-negative		−0.043***				−0.053***	−0.045***	−0.064***
		(0.00)				(0.00)	(0.00)	(0.00)
Vocal-positive			0.059***			0.076***	0.046***	0.101***
			(0.00)			(0.00)	(0.00)	(0.00)
Vocal-negative				−0.041***		−0.051***	−0.041***	−0.067***
				(0.00)		(0.00)	(0.00)	(0.00)
Verbal					0.014***	0.016***	0.014*	0.018***
					(0.00)	(0.00)	(0.08)	(0.00)
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Teacher fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Student fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,630	13,628	13,628	13,630	13,630	13,626	5,346	8,280

Notes. This table estimates how human teachers' emotions in the videos affect students' performances in subsequent games. The dependent variable is students' average move quality in each game, and the independent variables include measures of teachers' emotions in the revision classes before corresponding tournaments. Game controls include three variables capturing the average moving quality for the previous one, three, and seven games, respectively. Standard errors are clustered at the student and teacher levels. *p*-values are reported in parentheses.

***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6. AI Teacher's Emotional Status and Subsequent Student Performances

	Dependent variable							
	Average move quality							
Teacher	AI							
Student	Girls and boys						Girls	Boys
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Visual-positive	0.002					0.002	0.012	−0.006
	(0.68)					(0.70)	(0.13)	(0.48)
Visual-negative		0.000				0.000	0.007	−0.004
		(0.94)				(0.92)	(0.45)	(0.51)
Vocal-positive			−0.005			−0.005	−0.013	0.000
			(0.36)			(0.36)	(0.16)	(0.94)
Vocal-negative				0.001		0.001	−0.002	0.002
				(0.87)		(0.87)	(0.88)	(0.86)
Verbal					−0.002	−0.002	−0.009	0.003
					(0.77)	(0.76)	(0.57)	(0.72)
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Teacher fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Student fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Game controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,812	4,812	4,812	4,812	4,809	4,809	1,924	2,885

Notes. This table estimates how AI teacher's emotions in the video affect students' performances in subsequent games. The dependent variable is students' average move quality in each game, and the independent variables include measures of AI's emotions in the revision classes before corresponding tournaments. Game controls include three variables capturing the average moving quality for the previous one, three, and seven games, respectively. Standard errors are clustered at the student and teacher levels. *p*-values are reported in parentheses.

***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

From robustness checks presented in Online Appendix B.2, we discover that both male and female teachers exhibit discriminatory emotions.

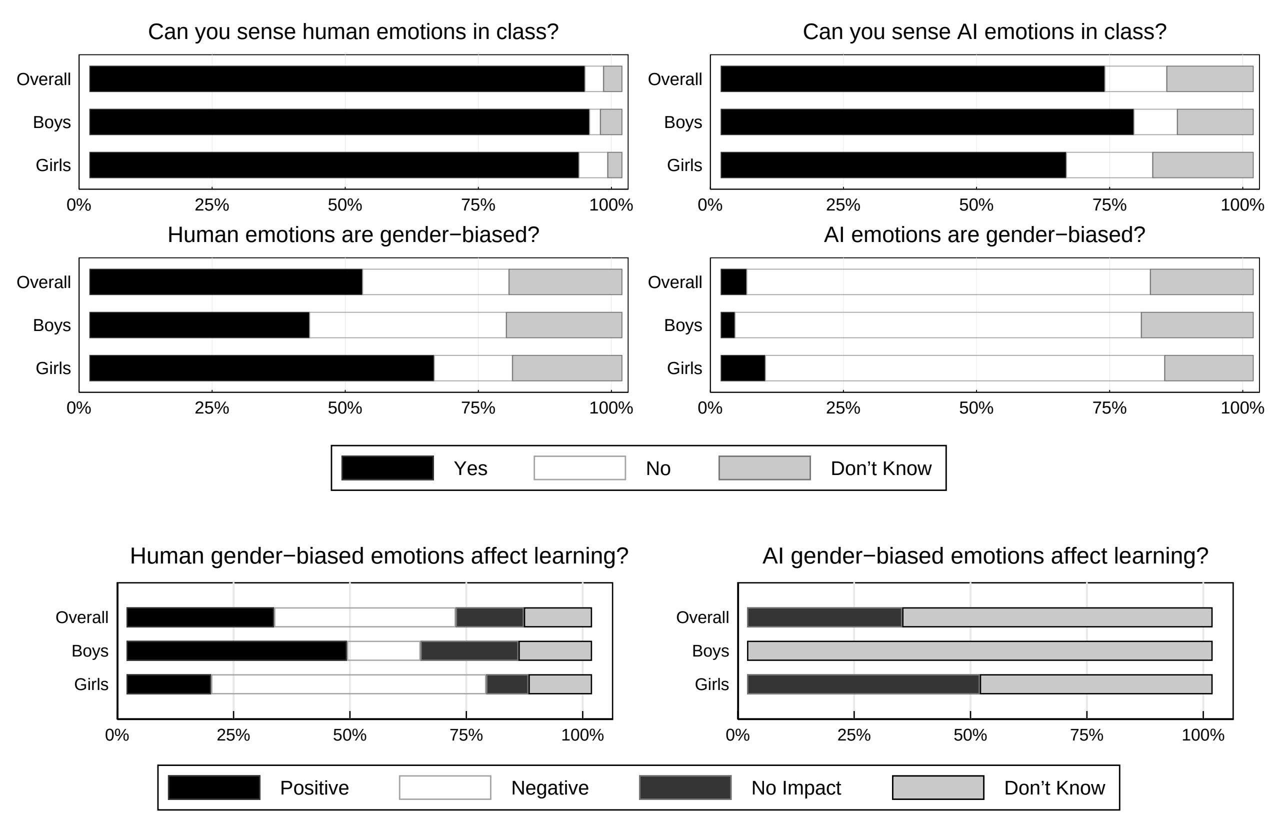
7.5.2. Teachers' Emotions and Students' Learning. We then explore whether teachers' emotions affect learning and regress students' average move quality on their teachers' emotion variables in the most recent review session. We first analyze the tournaments guided by human teachers.[24] Columns (1)–(6) of Table 5 show that human teachers' positive emotions enhance students' performance in the next tournament, and their negative emotions handicap students' performance in the subsequent game. The impacts of human teachers' emotions on students' performance are still valid when we perform regressions for boys and girls separately in columns (7) and (8), respectively.[25] The Blinder–Oaxaca decomposition reported in Online Appendix D finds that teachers' discriminatory emotional states can explain 27%–46% of the preexisting gender gap.

We then repeat the analyses for AI training in Table 6 and find that the AI teacher's emotions did not significantly affect students' future performance.

7.5.3. How Teachers' Gender-Biased Emotions Affect the Gender Gap. To examine the differential effects of teachers' biased emotions on boys and girls, we surveyed students to assess their ability to perceive human teachers' emotions. The upper left panel of Figure 8 illustrates that 93.9% of boys and 91.9% of girls responded positively. Subsequently, for those who acknowledge this ability, we inquired about their perception of gender bias in teachers' emotions. The middle-left panel shows that 41.3% of boys and 64.7% of girls perceived gender bias with a significant gender difference ($p = 0.06$). Furthermore, when we explored how these biased emotions influenced learning, the lower left panel indicated that 15.8% of boys and 59.1% of girls felt teachers' gender-biased emotions negatively impacted their progress with a significant gender difference ($p < 0.01$).

Next, we conducted similar analyses for the AI teacher. The upper right panel of Figure 8 demonstrates that 77.6% of boys and 64.9% of girls perceived the AI teacher's emotions. When asked about their perception of gender bias in the AI's emotions, the middle-right panel revealed that 2.6% of boys and 8.3% of girls perceived such bias with an insignificant gender difference

Figure 8. Survey Results on Students' Perception of Teachers' Emotions



Notes. The upper and middle panels of the figure show survey results on whether students can sense teachers' emotions and whether students believe these emotions are gender biased, conditional on being able to sense teachers' emotions, respectively. The lower panel of the figure reveals how teachers' gender-biased emotions affect learning, conditional on students believing teachers' emotions are gender biased. The detailed survey questions are listed as Q14 to Q19 from Online Appendix F.1.

(p = 0.40). Moreover, among those who believed in the AI's gender-biased emotions, none suggested that the AI's gender-biased emotions negatively affected their learning (lower right panel of the figure).

7.5.4. How Teachers' Emotions Affect Learning. To further investigate the impact of human teachers' emotions on learning, we surveyed students about the relationship between teachers' emotions and students' confidence, nervousness, concentration, willingness to take advice, and learning interest. We selected these five aspects following the literature on how teachers' emotions can influence students' performance (Yeager and Dweck 2012, Freeman et al. 2014). Specifically, we asked whether teachers' positive and negative emotions can intensify these five aspects. Students were required to choose one of three options (yes, no, don't know) for each question regarding if they can sense human teachers' emotions in class.

We present the survey results in Figure 9. In the two questions related to students' confidence, 87.5% reported that positive emotions from teachers boosted their confidence, whereas 62.5% believed that negative emotions did not increase their confidence. Regarding nervousness, 68.8% thought that teachers' positive emotions did not heighten their nervousness, and 85.0% stated that teachers' negative emotions increased their nervousness. In the question about students' concentration, 86.3% claimed that teachers' positive emotions helped them focus in class, whereas 65.0% responded that negative emotions did not improve their concentration. Concerning willingness to follow teachers' advice, 77.5% indicated that teachers' positive emotions made them more likely to comply compared with 13.8% for negative emotions. In the question about learning interest, 72.5% reported that positive emotions from teachers enhanced their interest, whereas only 11.3% said the same for negative emotions. In terms of gender differences, we found

Figure 9. Survey Results on How Teachers' Emotions Affect Students' Performance

Notes. This figure shows survey results on how teachers' positive and negative emotions affect students' confidence, nervousness, concentration, willingness to follow advice, and interest. The detailed survey questions are listed as Q20 to Q29 from Online Appendix F.1.

that both boys and girls responded similarly to the questions. Consequently, human teachers' gender-dependent emotional states (compared with girls, more positive and fewer negative emotions for boys) may contribute to boys being more confident, less nervous, more concentrated in class, more willing to follow advice, and more interested in learning, leading to the observed gender gap.

Result 1.5. Human teachers' positive emotions enhance students' confidence, attention, willingness to follow advice, and interest in classes, thereby improving learning outcomes; their negative emotions make students feel nervous and hamper learning.

8. Discussion

We evaluate the effectiveness of AI teaching through a natural experiment, in which AI replaced several *Go* teachers. The data demonstrate AI's advantages in improving education outcomes and reducing gender inequality. AI's strengths in game analysis, information provision, engaging presentation, and interactive display contribute to its superior performance compared with human teachers. Analysis of class recordings and surveys suggests that human teachers may display gender-biased emotions, potentially affecting girls negatively. AI teachers' display of gender-neutral emotions can help reduce this disparity.

We highlight the role of AI in online education, serving as both a valuable complement and substitute to human teachers, fostering a gender-neutral learning environment. With its widespread adoption across various sectors, AI has the potential to promote equality as evidenced in our study. Moreover, the data highlight the capacity of video-analyzing AI to detect discriminatory practices, expanding its utility beyond education. This information can aid writers, educators, and hiring committees in avoiding biased language and behaviors. Additionally, analyzing facial and vocal cues in business negotiations and legal proceedings may offer insights for managers and regulators.

However, our results only demonstrate the potential of AI; caution should be exercised when generalizing these findings to other AI-powered programs. To what extent we can extrapolate the above findings to other contexts depends on the field setting and the features of AI (Al-Ubaydli et al. 2017), such as AI capacity, the input of the AI, the nature of the task, and human–AI interactions.

Besides investigating the external validity of our findings, we envision several avenues for future research. First, enhancing AI teaching through interdisciplinary collaborations could yield significant improvements in learning outcomes for students with diverse learning and cognitive styles. Second, leveraging AI's information advantage to improve human-led teaching may be an intriguing study area.

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Endnotes

[1] Arizton predicts that the global board games market will reach 39.99 billion by 2028 from 18.93 billion in 2022 with a compound annual growth rate (CAGR) of approximately 13% during the forecast period. They also anticipate the AI-powered gaming market to hit 7.61 billion by 2026 from 1.56 billion in 2020, registering a CAGR of 30.0% during the forecast period. Interested readers can access the report at https://www.arizton.com/market-reports/global-board-games-market-industry-analysis-2024.

[2] Following the convention, we define the first 60 moves as the opening, the next 60 as the middle, and all moves afterward as the ending moves.

[3] The Elo rating system measures the relative strength of a player in zero-sum games. The higher a player's score, the more competitive the player is.

[4] The opponent may come from the same training agency if some students cannot find a pair from outside.

[5] The training agency's original plan was to switch back to the human teaching mode right after quarantined teachers returned. We contacted the agency about the idea of testing AI training rigorously and its potential benefit. In the end, they kindly implemented our experimental protocols, allowing the AI to continue teaching the treated group until the conclusion of the observation window.

[6] All observed variables are balanced across affected and unaffected students. Readers are referred to Table 1 for a balance check. It is also worth noting that students paid a nonrefundable tuition fee before the teaching year started, and we did not notice that any student in the treated group opted out of the program. The absence rate was less than 3% before and after the shock, and there were no significant differences in the attendance rate between the control and treated groups. Therefore, self-selection bias from voluntarily exiting the training program is unlikely to drive the main results. We provide detailed analyses of the attendance rate in Online Appendix A.7.

[7] One reason for focusing only on these two training semesters is that all classes were conducted online via Tencent Meeting software (Chinese equivalent of Zoom) because of COVID-19 preventative interventions in China (Bao and Huang 2021), giving us a clean comparison between human and AI teaching. The training agency mixed human teaching in both online and F2F formats with AI teaching after the sample period. F2F teaching differs from online teaching and may introduce additional confounding factors when

comparing human and AI teaching. In addition, the mixture of human- and AI-led classes makes it difficult to quantify the impact of AI teaching. Furthermore, the training agency did not record F2F classes, so we do not have data to analyze them.

[8] The KataGo program is available at https://github.com/lightvector/KataGo. We use KataGo (v1.11.0) to analyze all games played by students.

[9] One potential concern is that the KataGo algorithm may exhibit a bias toward understanding AI-trained moves and rating them higher. First, AI algorithms such as KataGo objectively evaluate moves based on effectiveness and efficiency in achieving victory rather than favoring AI-generated moves (Silver et al. 2016). Statistics from KataGo are also significantly correlated with the winning probability in our data. Second, researchers find the performance of AI-based *Go* algorithms is robust to diverse contexts and data distributions, suggesting KataGo's ability to evaluate moves for AI- and human-trained players (e.g., Wu 2019). Furthermore, Wu (2019) demonstrates KataGo's rating consistency across different sources, including other AI algorithms (e.g., AlphaGo, Leela Chess Zero) and human expert evaluations, ensuring objectivity in move evaluations.

[10] Some obstacles may prevent us from examining videos, including the slow internet connection, loud background noises, and poor light conditions, etc. We address the potential concern of sample selection bias by studying the determinant of video availability on characteristics of teachers and students in Online Appendix A.1. We show that the omission of videos is unrelated to any characteristics as all coefficients are statistically insignificant across different specifications.

[11] We use a composite negative emotion measure rather than using five variables to capture each negative emotion because machine learning algorithms are sometimes inaccurate in classifying subtly negative visual emotions, whereas combining negative emotions mitigates this concern (Sun et al. 2019).

[12] The Python package pyAudioAnalysis can be accessed at https://github.com/tyiannak/pyAudioAnalysis.

[13] The Baidu speech-to-text algorithm can be accessed at https://ai.baidu.com/tech/speech.

[14] The Hedonometer semantic analysis dictionary can be accessed at https://hedonometer.org/words/labMT-zh-v2.

[15] The measure of the verbal dimension only has one variable, which is different from the measurements for visual and vocal dimensions. This difference, however, does not create inconsistency as the score from the verbal dimension is positively correlated with the positive emotion measurement of the other two dimensions and negatively correlated with the negative emotion measurement of the other two dimensions.

[16] The underlying assumption for the DID model is that outcomes for students in treated and control groups would have maintained a parallel trend in the absence of AI training (i.e., experimental intervention drives the differences between the treated and control groups). This parallel trend assumption is likely valid as students' performance before the experimental intervention between control and treated groups is similar.

[17] In Online Appendices A.2 and A.3, we show that our results are robust to alternative fixed effects specifications and standard error clustering.

[18] This result is robust when we consider the improvement in average move quality in different game stages, including opening, middle, and ending as shown in Online Appendix C.1.

[19] As mentioned in Section 4.2, the average move quality variable collects all move information in a game and compares them against a common standard. Therefore, it is a comprehensive variable approximating students' skills. In addition, it is a continuous variable, which can avoid computational difficulties associated with processing discrete dependent variables. Therefore, we treat the average move quality as the primary measure for students' learning outcome. Our results are robust when we use different measures, including the number of errors/critical errors and magnitude of errors as shown in Online Appendix C.2.

[20] One concern that could limit the scope of this paper is the potential ineffectiveness of online training in comparison with F2F training coupled with its limited popularity in educational settings. This raises the possibility of overestimating the effectiveness of AI training. Whereas our survey reveals that more than half of the students consider F2F teaching superior to online teaching, the literature suggests that the market for online teaching is expanding, and its effectiveness is on the rise. These points are further detailed in Online Appendix C.5.

[21] Boys may have lower self-control than girls, especially in AI-led classes, potentially disadvantaging their AI training. However, this explanation is unlikely: survey responses on absent-mindedness in human- and AI-led classes reveal no significant gender-based variance. We refer readers to Online Appendix A.6 for more details.

[22] Without further specification, all p-values comparing survey answers are obtained from two-sample Mann–Whitney tests.

[23] One concern about the relationship between teachers' emotions and students' gender is that the gender variable may contain unobserved information regarding student quality. Therefore, the omitted quality proxies rather than gender discrimination may drive teachers' emotional changes. We address this possibility using the test developed by Oster (2019). Interested readers are referred to further results in Online Appendix B.1.

[24] There was no revision session before the first tournament of the observation window. However, all teachers gave an opening remark before the first training session to welcome their students and announce the course-related specifics and house rules. We use the video information of these opening speeches to explain students' performance in their first match. All results remain robust if we remove these observations.

[25] It is worth noting that all coefficients are standardized; therefore, we cannot compare the size of coefficients in columns (7) and (8) to study whether boys and girls have different sensitivity to teachers' emotions. In Online Appendix B.3, we interact each emotion variable with students' gender and find no gender difference in sensitivity to teachers' emotions.

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